

Exam Date & Time: 04-May-2024 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

SECOND SEMESTER B.TECH. DEGREE EXAMINATIONS - APRIL / MAY 2024

SUBJECT: PHY 1071-PHY/ PHY 1071-B/ PHY 1051-B - ENGINEERING PHYSICS

ENGINEERING PHYSICS [PHY 1071-PHY]

Marks: 50

Duration: 180 mins.

A

Answer all the questions.

- 1A) i) Write the physical significance of Gauss law in electricity and magnetism. (5)
ii) Explain the working principle of Michelson's interferometer with a diagram.
- 1B) Light from an argon laser strikes a diffraction grating that has 5310 grooves per centimeter. The central and first-order principal maxima are separated by 0.488 m on a wall 1.72 m from the grating. Determine the wavelength of the laser light. (3)
- 1C) Sketch probability density curves for the lowest energy state of the particle in an infinite potential well and finite potential well. (2)
- 2A) Obtain the expression for energy density of radiation in terms of Einstein Coefficients. Compare it with Planck's radiation equation and hence show that the probability of induced absorption is equal to probability of stimulated emission. (5)
- 2B) A glass clads with core glass of refractive index 1.5 and cladding is doped to give a fractional refractive index change of 0.0005. Determine (a) the refractive index of the cladding (b) the critical angle (c) the numerical aperture. (3)
- 2C) For copper at 300 K, calculate the probability that a state with an energy equal to 99.0% of the Fermi energy is occupied. Fermi energy of copper is 7.05 eV. (2)
- 3A) Using the appropriate diagram, derive the expression for the wavelength [Compton] shift for electromagnetic radiation such as X-rays, interacting with a free or nearly free electron. (5)
- 3B) The stopping potential for photoelectrons released from a metal is 1.48 V larger than in another metal. If the threshold frequency for the first metal is 40.0 % smaller than the second metal, determine the work function for each metal. (3)
- 3C) An FM radio transmitter has a power output of 150 kW and operates at a frequency of 99.7 MHz. How many photons per second does the transmitter emit? (2)
- 4A) A particle possessing energy E is confined within a finite potential well of height U (where $U > E$) with the following potential function: (5)

$$U(x) = 0 \text{ for } 0 < x < L$$

$$U(x) = U \text{ for } x \leq 0, x \geq L.$$

Using the appropriate diagram, write the Schrödinger equation and the wavefunctions for all three regions ($x < 0$, $x > L$, and $0 < x < L$). Also, write down the boundary conditions applicable to the wave function at $x = 0$ and $x = L$.

- 4B) Find the wavelength of the incident photon needed to move the electron from the ground state to the third excited state in a metal with a length of 10^{-10} meters. (3)

- 4C) What is the minimum frequency of a photon required to elevate an electron from the valence band to the conduction band in silicon at 300 Kelvin, given an energy gap of 1.14 eV? (2)

- 5A) Using the expression for quantized energy levels for electron in a three-dimensional box of side L , obtain an expression for density of states in a metal. [Given: For a three-dimensional box (cube) of side L , energy eigen values are given by:

$$E_n = \frac{\hbar^2 \pi^2}{2m_e L^2} (n_x^2 + n_y^2 + n_z^2). \quad (5)$$

- 5B) The $J=0$ to $J=1$ rotational transition of the CO molecule occurs at a frequency of 1.15×10^{11} Hz. Use this information to calculate, moment of inertia and bond length of the molecule. Given: Atomic masses of carbon and oxygen are 12 u and 16 u respectively. 1 (3)
 $u = 1.67 \times 10^{-27}$ kg.

- 5C) Surface to volume ratio decreases with decreasing size of the spherical particles. State whether this statement is true or false. Justify your answer. (2)

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